Welcome!

HEXAPOD WITH SENSORS

Market analysis

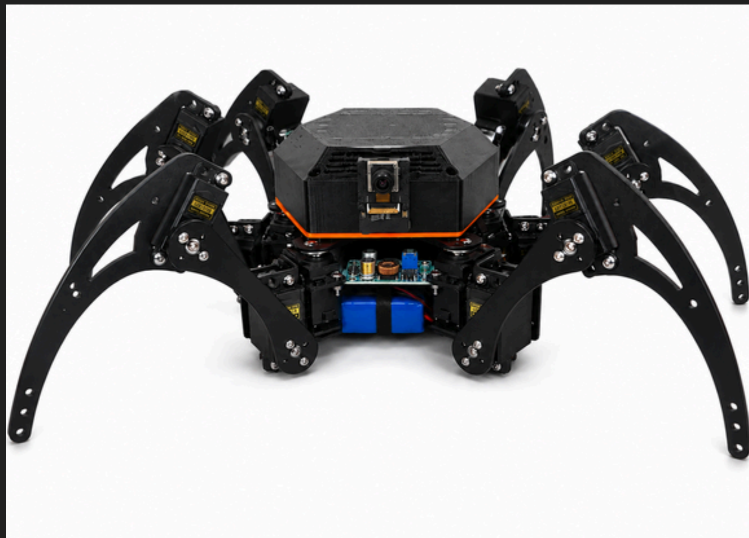
The global market for hexapod robots is growing rapidly across defense, industrial inspection, agriculture, healthcare, research, and disaster response, driven by advancements in robotics, IoT, AI, and automation.

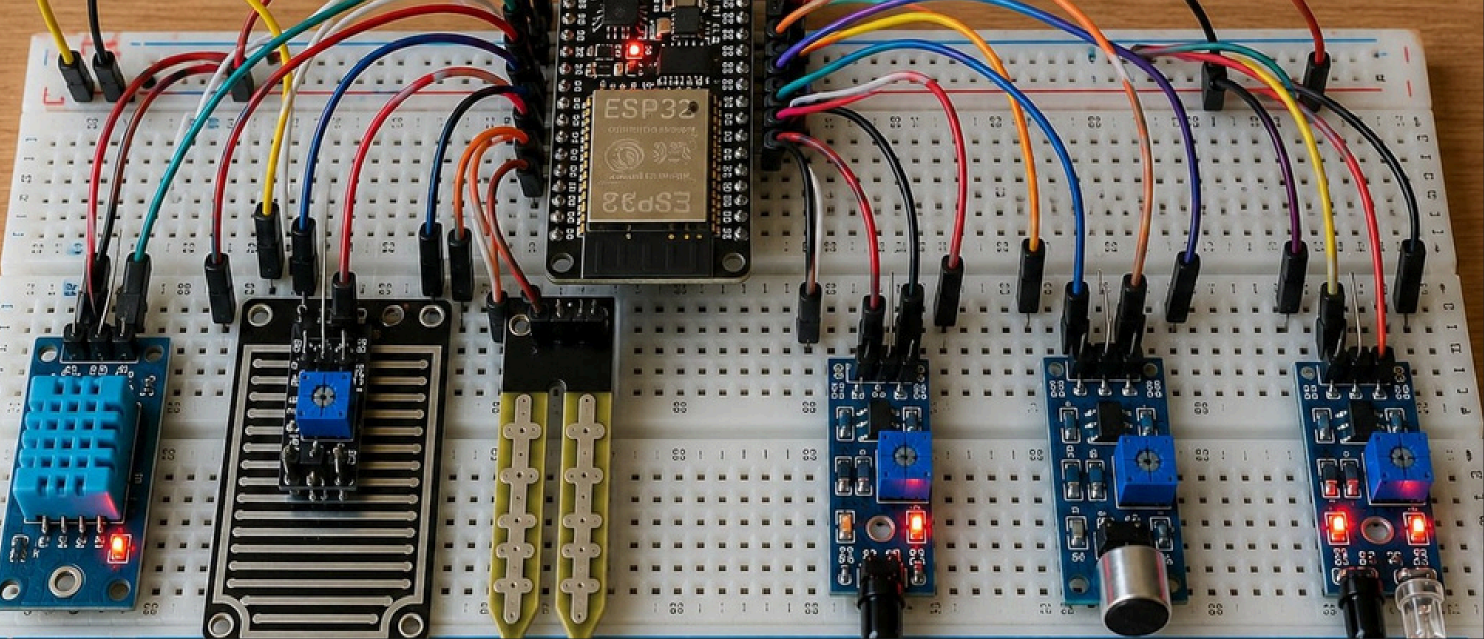
Project goals and objectives

This project presents a Hexapod Robot built using an ESP32 microcontroller and an 18-servo locomotion system. The robot is designed to achieve stable movement using a six-legged walking mechanism, making it suitable for navigating uneven terrain while maintaining balance.

Implementation strategy

The hexapod robot was implemented by designing the mechanical structure, interfacing 18 servo motors with an ESP32 through PCA9685 servo drivers, and integrating sensors for real-time data collection. A Bluetooth-based mobile app was developed for robot control





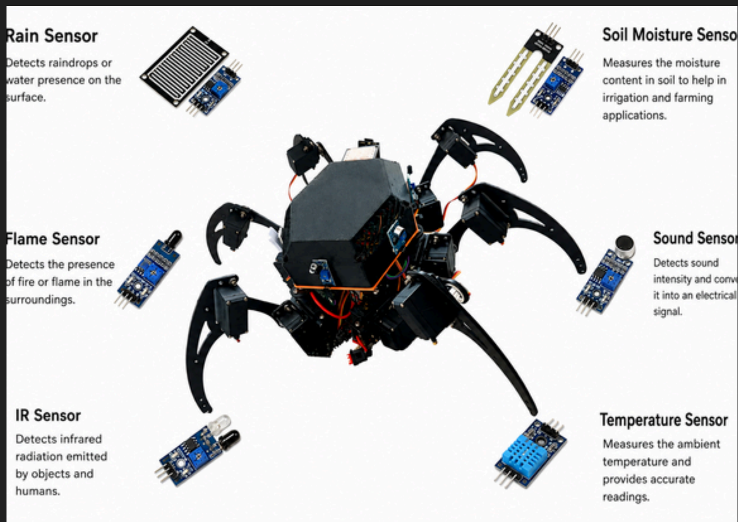
Sensors

- Temperature Sensor 🌡️
- Rain Sensor 🌧️
- Soil Moisture Sensor 🌱
- Flame Sensor 🔥
- Sound Sensor 🎤
- IR (Infrared) Sensor 📡

Creation of Strategic plan

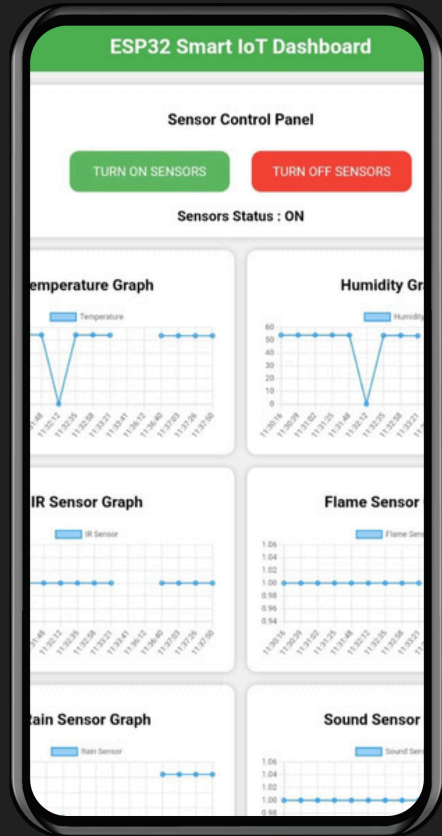
The sensor integration strategy focused on placing six environmental sensors around the hexapod robot to achieve comprehensive 360° surroundings monitoring. Temperature, rain, soil moisture, flame, sound, and IR sensors were interfaced with the ESP32.

HexaGLAM_K



The App named by HexaGLAM_K created using mit app to control the hexapod robot with to monitor the sensors data with the graphs using Thingspeak and getting in app

HexaGLAM_K App



THANKS

for your assistance

www.iiitk.ac.in/
www.trigunrobotics.com